Excretory Products and Their Elimination

Topic – Full Chapter

1. Given below are two statements :

Statement-I: Nephridia are the tubular excretory structures of earthworms.

Statement-II: Malpighian tubules help in the removal of nitrogenous wastes and osmoregulation.

Choose the most appropriate answer

(1) Statement I and Statement II both are correct.

(2) Statement I is correct, but statement II is incorrect.

(3) Statement I is incorrect, but Statement II is correct.

(4) Statement I and Statement II both are incorrect.

2. Which one is incorrect about mechanism of conentration of filtrate ?

(1) NaCl is transported by the ascending limb of 'Loop of Henle' which is the exchanged with the descending limb of vasa-recta.

(2) Help in maintaining an increasing osmolarity towards the inner medullary interstitum.

(3) The proximity between the 'Loop of Henle' and vasa recta as well as counter current in them help in maintaining the osmolarity of medullary interstitium.

(4) Small amount of urea enter in the thick segment of ascending limb of 'Loop of Henle' which is transported back by collecting duct.

3. Tick mark the correct option ?

(1) Aquatic amphibians → Ammonotelic.

(2) Terrestrial amphibians → Uricotelic

(3) Marine fishes → Ammonotelic

(4) Land snail → Ureotelic.

4. We have to observe that glucose and ketone bodies pass through urine in a person. Such condition indicates sign of :

(1) Diabetes insipidus

(2) Glomerulonephritis

(3) Diabetes mellitus

(4) Renal failure.

5. Match List-I with List-II

List-I	List-II
I. Length of kidney	**A. 10-12 cm**
II. Width of kidney	**B. 5-7 cm**
III. Thickness of kidney	**C. 120-170 gram**
IV. Weight of kidney	**D. 2-3 cm**

(1) I-A, II-B, III-D, IV-C

(2) I-C, II-D, III-B, IV-A

(3) I-B, II-A, III-D, IV-C

(4) I-C, II-D, III-A, IV-B

6. First step of urine formation is;

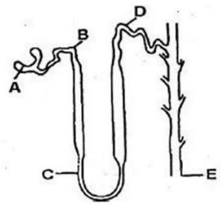
(1) Ultrafiltration

(2) Tubular secretion

(3) Selective secretion

(4) Tubular reabsorption

7. Use following diagram to complete the statements about the human nephron-



I. The composition of the filtrate would be most like plasma in the tubule next to the letter.

II. The urine would be most concentrated in the collecting duct next to letter.

III. Most of the glomerular filtrate is reabsorbed into peritubular capillary next to the letter

IV. Conducting of urine to pelvis of the kidney from the structure next to the letter

V. Most water is reabsorbed by the structure next to the litter

(1) I-A, II-C, III-B, IV-E, V-D

(2) I-A, II-E, III-B, IV-C, V-D

(3) I-A, II-B, III-E, IV-C, V-D

(4) I-A, II-E, III-B, IV-E, V-B

8. The part of nephron that opens into collecting duct is;

(1) Distal convoluted tubule (DCT)

(2) Proximal convoluted tubule (PCT)

(3) Henle's loop (HL)

(4) Glomerulus

9. Damage in filtering part of nephron is known as;

(1) Glomerulonephritis

(2) Renal calculi

(3) Kidney stone

(4) Tubular damage

10. Given below are two statements :

Assertion(A): Human kidneys can produce urine nearly four times concentrated than initial filtrate formed.

Reason(R): On an average, 25-30 gm of urea is excreted out per day.

(1) Both Assertion (A) and Reason (R) are true, and Reason (R) is a correct explanation of Assertion (A).

(2) Both Assertion (A) and Reason (R) are true, but Reason (R) is not a correct explanation of Assertion (A).

(3) Assertion (A) is true, and Reason (R) is false.

(4) Assertion (A) is false, and Reason (R) is true.

11. Which statement is false?

(1) Angiotensin II, being a powerful vasoconstrictor, increases glomerular pressure and thereby GFR

(2) Angiotensin II activates the adrenal cortex to release aldosterone

(3) Aldosterone promotes reabsorption of Na+ and water from the DCT and CT leading to an increase in B.P. and GFR

(4) ANF causes vasoconstriction

12. ADH is synthesised by, ______ released by _____ and acts on ________

(1) Hypothalamus, Neurohypophysis, DCT and CT

(2) Hypothalamus, Neurohypophysis, Loop of Henle

(3) Hypothalamus, Adenohypophysis, DCT and CT

(4) Hypothalamus, Adenohypophysis, Loop of Henle

13. Which is the correct pathway for passage of urine in humans?

(1) Collecting tubule → ureter → bladder → urethra

(2) Renal vein → renal ureter → bladder → urethra

(3) Pelvis→ Medulla→ bladder→ urethra

(4) Cortex→ Medulla→ bladder→ ureter

14. Consider the following statements

Statement-I: Dialysing fluid having the same composition as that of plasma including the nitrogenous wastes.

Statement-II: Haemodialysis is a boon for thousands of uremic patients all over the world.

Choose the most appropriate answer

(1) Statement I and Statement II both are correct.

(2) Statement I is correct, but statement II is incorrect.

(3) Statement I is incorrect, but Statement II is correct.

(4) Statement I and Statement II both are incorrect.

15. Which of the following is correctly matched?

(1) Afferent arterioles – Carry blood away from glomerulus towards renal vein.

(2) Podocytes – Create slit pores for filtration of blood into the Bowman's capsule.

(3) Henle's loop – Most reabsorption of major substances from filtrate.

(4) DCT – Reabsorption of K+ ions into surrounding blood capillaries

16. How many statements are correct ?

(a) vasa recta is absent or highly reduced in cortical nephrons

(b) ammonia is less toxic as compared to urea and requires large amount of water for its elimination

(c) GFR in a healthy individual is approximately 125 ml / minute

(d) PCT also help to maintain the pH and ionic balance of the body fluid

(1) one

(2) two

(3) three

(4) four

17. Match List-I with List-II

List-I	List-II
I. Antennal glands	**A. Protonephridia**
II. Urea cycle	**B. Kidney**
III. Platyhelminthes	**C. Prawn**
IV. Vertebrates	**D Liver**

(1) I-A, II-B, III-D, IV-C

(2) I-C, II-D, III-B, IV-A

(3) I-B, II-A, III-D, IV-C

(4) I-C, II-D, III-A, IV-B

18. Which of the following is correct about hilum of kidney?

(1) It is present at convex outer surface.

(2) It is present at inner convex surface.

(3) It is a notch through which ureter, nerve and blood vessel enter.

(4) It is a place where calyces open.

19. Given below are two statements :

Assertion(A): Urinary bladder and ureters are lined by transitional epithelium.

Reason(R): The transitional epithelium in the ureter and urinary bladder allows for stretching to accommodate urine without getting torn.

(1) Both Assertion (A) and Reason (R) are true, and Reason (R) is a correct explanation of Assertion (A).

(2) Both Assertion (A) and Reason (R) are true, but Reason (R) is not a correct explanation of Assertion (A).

(3) Assertion (A) is true, and Reason (R) is false.

(4) Assertion (A) is false, and Reason (R) is true.

20. Following are the steps of dialysis

A. Blood is passed into a vein.

B. Blood is mixed with heparin.

C. Blood is mixed with anti-heparin.

D. Blood is drained from convenient artery.

E. Blood is passed through a coiled and porous cellophane tube bathing in dialysis fluid.

F. Removal of nitrogenous wastes from blood.

The correct sequence of steps is-

(1) $A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow F$

(2) $F \rightarrow C \rightarrow E \rightarrow B \rightarrow A \rightarrow D$

(3) $D \rightarrow B \rightarrow E \rightarrow F \rightarrow C \rightarrow A$

(4) $D \rightarrow C \rightarrow E \rightarrow F \rightarrow B \rightarrow A$

ANSWER KEY

1. (1) 2. (4) 3. (1) 4. (3) 5. (1) 6. (1) 7. (4) 8. (1) 9. (1) 10. (2)

11. (4) 12. (1) 13. (1) 14. (3) 15. (2) 16. (3) 17. (4) 18. (3) 19. (1) 20. (3)